

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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1. TITLE

Design Static Webpage using HTML Components

2. AIM

To create a static webpage using HTML elements such as headings, lists, links, images, tables, and forms without using external CSS or JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the structural components of an HTML5 document.
 - To learn how to use HTML heading tags (<h1> to <h6>) and list tags (, ,) for organizing content.
 - To implement hyperlinks using anchor tags (<a>) and load images using tags.
 - To construct structured data tables using <table>, <tr>, <th>, and <td> tags.
 - To design interactive user forms using <input>, <select>, <textarea>, <input type="submit">, and <input type="reset"> controls.
 - To analyze the visual rendering of block and inline HTML components under browser defaults.
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4. THEORY

HTML (HyperText Markup Language) is the standard markup language for documents designed to be displayed in a web browser. It forms the core structure of all web pages. Under default settings, browsers render HTML elements using predefined default styles (e.g., standard margins, blue underlined links, serif fonts). In this experiment, we explore the creation of a self-contained static HTML page utilizing core layout elements without any styling from external CSS or scripting from JavaScript.

Theory of HTML Components:

- **Document Structure:** The <!DOCTYPE html> declaration identifies the document as an HTML5 document. The <html> element is the root. The <head> contains metadata (character set, viewport, page title) and <body> contains visible content.
- **Structural Sections:** HTML5 semantic containers like <header>, <nav>, <main>, <section>, and <footer> logically partition layout sections.
- **Headings & Paragraphs:** Headings (<h1>-<h6>) establish hierarchical structure; paragraphs (<p>) contain text blocks.

- **Lists & Links:** Lists (`<ul>`, `<ol>`, `<li>`) organize text items; anchors (`<a>`) create local and external hyperlinks.
 - **Tables:** `<table>` structures data grid-style using rows (`<tr>`), header columns (`<th>`), and data cells (`<td>`).
 - **Forms:** `<form>` collects user inputs using text/email fields, dropdown selection selectors, and multi-line textareas.
 - **Block vs. Inline:** Block elements (e.g., `<p>`, `<table>`, `<form>`) take full width starting on new lines; inline elements (e.g., `<a>`, `<img>`) fit the inline text flow.
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5. ALGORITHM / PROCEDURE

1. Create project directories: Record/Experiment - 1(week -1)/Src and Documents.
 2. Create Index.html in the Src directory.
 3. Structure the file using the HTML5 doctype declaration and basic tags (html, head, body).
 4. Configure metadata in the `<head>` (UTF-8, responsive viewport) and define the document title.
 5. Define semantic structure: `<header>`, navigation `<nav>`, main container `<main>`, data `<section>`, and `<footer>`.
 6. Implement internal anchor links (`#home`, `#about`, `#services`, `#contact`).
 7. Construct a structured layout table for services and pricing using `<table>`, `<thead>`, `<tbody>`, `<tr>`, `<th>`, and `<td>`.
 8. Design a user contact form requiring name, email, inquiry reason dropdown, message textarea, and submission button.
 9. Run the file locally in Google Chrome and verify all functionalities (navigation, validation, clear form).
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6. PROGRAM

Index.html - Part 1 (Lines 1-54)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Static Webpage</title>
</head>
<body>

  <!-- Header & Navigation -->
  <header>
    <h1>My Website Header</h1>
    <nav>
      <a href="#home">Home</a> |
      <a href="#about">About</a> |
      <a href="#services">Services</a> |
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <hr>

  <!-- Main Content Area -->
  <main>

    <!-- Home Section -->
    <section id="home">
      <h2>Welcome to My Webpage</h2>
      <p>This is a static webpage built purely using HTML tags without external CSS or JavaScript.</p>
    </section>

    <hr>

    <!-- About Section -->
    <section id="about">
      <h2>About Us</h2>
      <p>Here is a list of features included on this site:</p>
      <ul>
        <li>Semantic HTML structure</li>
        <li>Organized content sections</li>
        <li>Native HTML input forms</li>
        <li>Data structured with tables</li>
      </ul>
    </section>

    <hr>

    <!-- Services Section -->
    <section id="services">
      <h2>Our Services & Pricing</h2>
      <table border="1" cellpadding="8" cellspacing="0">
        <thead>
          <tr>
            <th>Service</th>
            <th>Description</th>
          </tr>
        </thead>
      </table>
    </section>
  </main>
</body>
</html>
```

Index.html - Part 2 (Lines 55-109)

```
        <th>Price</th>
    </tr>
</thead>
<tbody>
    <tr>
        <td>Web Design</td>
        <td>Creating clean static layouts.</td>
        <td>$100</td>
    </tr>
    <tr>
        <td>HTML Coding</td>
        <td>Semantic HTML5 page structuring.</td>
        <td>$75</td>
    </tr>
</tbody>
</table>
</section>

<hr>

<!-- Contact Form Section -->
<section id="contact">
    <h2>Contact Us</h2>
    <form action="#" method="post">
        <label for="name">Name:</label><br>
        <input type="text" id="name" name="name" required><br><br>

        <label for="email">Email:</label><br>
        <input type="email" id="email" name="email" required><br><br>

        <label for="subject">Subject:</label><br>
        <select id="subject" name="subject">
            <option value="general">General Inquiry</option>
            <option value="support">Support</option>
            <option value="feedback">Feedback</option>
        </select><br><br>

        <label for="message">Message:</label><br>
        <textarea id="message" name="message" rows="5" cols="30" required></textarea><br><br>

        <input type="submit" value="Submit Form">
    </form>
</section>

</main>

<hr>

<!-- Footer -->
<footer>
    <p>&copy; 2026 My Static HTML Page. All rights reserved.</p>
</footer>

</body>
</html>
```

7. CODE EXPLANATION

HTML Tag	Explanation
<!DOCTYPE html>	Declares the document as an HTML5 document.
<html>	Root element of the HTML document containing all other elements.
<head>	Contains document metadata, character encoding, and page title.
<title>	Specifies the title displayed in the browser tab.
<body>	Contains all the visible content displayed on the webpage.
<header>	Defines introductory header content (e.g. main titles, subheadings).
<nav>	Defines a block containing navigation links.
<a>	Renders a hyperlink targeting URLs or local elements via href.
<hr>	Inserts a thematic break or horizontal divider between layout sections.
<main>	Encapsulates the central, dominant content of the document body.
<section>	Defines a semantic standalone section within the page.
<h1>, <h2>	Heading elements establishing clear structural hierarchy.
<p>	Represents paragraph blocks of descriptive text.
&	Represent unordered bulleted lists and their individual items.
<table>	Constructs data tables using rows and structured cells.
<thead> & <tbody>	Partitions table header rows and table body data rows.
<tr>, <th>, <td>	Define table rows, bold header cells, and standard data cells.
<form>	Defines an interactive user form container for input controls.
<label>	Provides accessible captions and labels for form input elements.
<input>	Defines input controls (text, email, and submit button).
<select> & <option>	Constructs dropdown menus with selectable choices.
<textarea>	Creates multi-line text input fields for detailed messages.
<footer>	Defines footer elements containing author/copyright info.

8. TEST CASES AND EXECUTION DATA

The following test suite was executed manually in Google Chrome to verify HTML components and input validation:

TC ID	Test Description	Input / Action	Expected Result	Status
TC-01	Local Anchor Navigation	Click on '#about' in navigation	Browser smooth-scrolls and views the About Us section	Pass
TC-02	Service Table Navigation	Click on '#services' in navigation	Browser jumps directly to Our Services & Pricing table	Pass
TC-03	Required Field Validation	Leave 'Name' blank and click Submit	Form submission blocked; browser displays required field tooltip	Pass
TC-04	Email Address Format Validation	Enter 'invalidemail' in email field and submit	Form submission blocked; email format warning shown	Pass
TC-05	Reason Dropdown Selection	Select 'Feedback' in subject dropdown	Dropdown updates and retains selected value upon submission	Pass
TC-06	Table Data Rendering Integrity	Inspect Services table on screen	3 columns (Service, Description, Price) render with clean borders	Pass

9. OUTPUT

Output: Rendered Webpage in Chrome Browser

My Website Header

[Home](#) | [About](#) | [Services](#) | [Contact](#)

Welcome to My Webpage

This is a static webpage built purely using HTML tags without external CSS or JavaScript.

About Us

Here is a list of features included on this site:

- Semantic HTML structure
- Organized content sections
- Native HTML input forms
- Data structured with tables

Our Services & Pricing

Service	Description	Price
Web Design	Creating clean static layouts.	\$100
HTML Coding	Semantic HTML5 page structuring.	\$75

Contact Us

Name:

Email:

Subject:

General Inquiry ▼

Message:

Figure 1: Rendered static HTML webpage displaying header, navigation, content sections, pricing table, and contact form.

10. RESULT

Thus, a static webpage using various HTML components (headings, lists, tables, links, and forms) was successfully designed, implemented, and verified using manual test cases under browser defaults.

Submission Package Structure:

```
Experiment - 1(week -1)/
├── Documents/
│   ├── Experiment_1.docx
│   └── Experiment_1.pdf
├── Screenshots/
│   └── Screenshot 2026-08-16 222714.png
└── Src/
    └── Index.html
```